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FAN AND MANUFACTURING METHOD THEREOF

Abstract

A fan and a manufacturing method thereof are provided. The fan includes a hub having a circumference, a plurality of first blades, a mute ring and a plurality of second blades. Each of the plurality of first blades includes a first terminal disposed at the circumference of the hub and a second terminal opposite to the first terminal. The mute ring is disposed at the second terminal of each of the plurality of first blades. The plurality of second blades are disposed at the second terminal of each of the plurality of first blades and connected to the mute ring, wherein the plurality of second blades and the plurality of first blades are alternately arranged.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Application Ser. No. 63/561,592 filed on Mar. 5, 2024 and claims priority to China Patent Application No. 202411285365.3 filed on Sep. 13, 2024. The entire contents of the above-mentioned applications are incorporated herein by reference for all purposes.

FIELD OF THE INVENTION
[0002] The present disclosure relates to a fan and a manufacturing method thereof, and more particularly to a fan with improved structure and performance and the manufacturing method thereof.
BACKGROUND OF THE INVENTION
[0003] Currently, blades of fans in the notebook are generally divided into plastic and metal ones. Due to the mold and the injection molding process, there are limitations on the thickness and height of the plastic blades. As for the metal blades, the number thereof is limited due to the production process. Accordingly, no matter the plastic blade or the metal blade, the number thereof as used in the fan will be limited thereby, which also influences the performance of the fan.
[0004] Fans in current notebook also include mute ring. Generally, the mute ring is made of plastic material, and the thickness thereof is similarly limited by the mold and the injection molding process. For example, the least thickness is about 0.5 mm, which might reduce the working area of the blades and thus influence the performance of the fan.
[0005] Therefore, there is a need of providing a fan and a manufacturing method thereof for improving the drawbacks above.
SUMMARY OF THE INVENTION
[0006] An object of the present disclosure is to improve the performance of a fan through improving the material and/or the structure thereof.
[0007] Another object of the present disclosure is to provide a fan which employs metal material to form a mute ring for reducing the thickness thereof, so as to achieve the effect of improving the performance of fan.
[0008] A further object of the present disclosure is to provide a fan and a manufacturing method thereof, wherein the performance of fan is improved by adding blades on existing blades of the fan without altering the original manufacturing process, which is easy to conduct with significant effects.
[0009] In accordance with an aspect of the present disclosure, a fan is provided. The fan includes a hub having a circumference, a plurality of first blades, a mute ring and a plurality of second blades. Each of the plurality of first blades includes a first terminal disposed at the circumference of the hub and a second terminal opposite to the first terminal. The mute ring is disposed at the second terminal of each of the plurality of first blades. The plurality of second blades are disposed at the second terminal of each of the plurality of first blades and connected to the mute ring, wherein the plurality of second blades and the plurality of first blades are alternately arranged.
[0010] In an embodiment, each of the plurality of second blades includes a third terminal toward the hub and a fourth terminal opposite to the third terminal, and a distance between the third terminal and the fourth terminal of each of the plurality of second blades is less than a perpendicular distance between the second terminal of each of the plurality of first blades and the circumference.

[0011] In an embodiment, the plurality of first blades are made of a metal material, and the plurality of second blades are made of a plastic material.
[0012] In an embodiment, the mute ring is made of the plastic material and is integrally formed with the plurality of second blades.
[0013] In an embodiment, the mute ring includes a metal base and a plastic outer portion, and the plastic outer portion is integrally formed with the plurality of second blades.
[0014] In an embodiment, the mute ring is stabilized at the second terminal of each of the plurality of first blades through the plastic outer portion thereof.
[0015] In an embodiment, the mute ring is stabilized through connecting the metal base thereof to the second terminal of each of the plurality of first blades.
[0016] In accordance with another aspect of the present disclosure, a manufacturing method of a fan is provided. The method includes steps of (a) providing a hub having a circumference; (b) providing a plurality of first blades, each of the plurality of first blades including a first terminal and a second terminal opposite to the first terminal, and disposing the first terminal of each of the plurality of the first blades at the circumference of the hub; and (c) performing an injection molding process to form a mute ring and a plurality of second blades at the second terminal of each of the plurality of first blades, wherein the plurality of second blades and the plurality of first blades are alternately arranged.
[0017] In an embodiment, each of the plurality of second blades includes a third terminal toward the hub and a fourth terminal opposite to the third terminal, and a distance between the third terminal and the fourth terminal of each of the plurality of second blades is less than a perpendicular distance between the second terminal of each of the plurality of first blades and the circumference.
[0018] In an embodiment, the plurality of first blades are made of a metal material, and the plurality of second blades are made of a plastic material.
[0019] In an embodiment, the mute ring is made of the plastic material and is integrally formed with the plurality of second blades.
[0020] In an embodiment, the mute ring includes a metal base and a plastic outer portion.
[0021] In an embodiment, the step (c) further includes steps of: (c1) disposing the metal base at the second terminal of each of the plurality of first blades; and (c2) performing the injection molding process to form the plastic outer portion and the plurality of second blades.
[0022] In an embodiment, the step (c1) further includes: positioning the metal base at the second terminal of each of the plurality of first blades through a tool.
[0023] In an embodiment, the step (c1) further includes: connecting the metal base to the second terminal of each of the plurality of first blades.
[0024] In accordance with another aspect of the present disclosure, a fan is provided. The fan includes a hub having a circumference, a plurality of first blades made of a metal material, and a mute ring made of a plastic material. Each of the plurality of first blades includes a first terminal disposed at the circumference of the hub and a second terminal opposite to the first terminal, and the mute ring is disposed at the second terminal of each of the plurality of first blades.
[0025] In an embodiment, the second terminal of each of the plurality of first blades is bent to form a soldering surface, the mute ring includes a plurality of openings respectively corresponding to the soldering surface of the second terminal of each of the plurality of first blades, and the mute ring is fixed through soldering the soldering surface of the second terminal of each of the plurality of first blades respectively with each of the plurality of openings.
[0026] In an embodiment, the second terminal of each of the plurality of first blades comprises a first engaging structure, the mute ring comprises a plurality of second engaging structures respectively corresponding to the first engaging structure of the second terminal of each of the plurality of first blades, and the mute ring is fixed through engaging the plurality of second

engaging structures respectively with the first engaging structure of the second terminal of each of the plurality of first blades.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] The above contents of the present disclosure will become more readily apparent to those ordinarily skilled in the art after reviewing the following detailed description and accompanying drawings, in which:

[0028] FIG. **1** is a schematic view showing a fan according to a first embodiment of the present disclosure;

[0029] FIG. **2** is a comparison table showing data of a fan adopting a metal mute ring according to the present disclosure and a control fan with non-metal mute ring;

[0030] FIG. **3**A is a schematic view showing a fan according to a second embodiment of the present disclosure;

[0031] FIG. **3**B is a schematic view showing the fan according to the second embodiment of the present disclosure from another view angle;

[0032] FIG. **4**A is an enlarged view of square A in FIG. **3**A;

[0033] FIG. **4**B is an enlarged view of square B in FIG. **3**B;

[0034] FIG. **5** is a top view of the fan according to the second embodiment of the present disclosure;

[0035] FIG. **6** is a graph comparing performances of a fan adopting metal first blades and plastic second blades and a fan adapting metal blades only;

[0036] FIG. **7** is a flow chart illustrating a manufacturing method of a fan according to an embodiment of the present disclosure;

[0037] FIG. **8** is a flow chart illustrating a manufacturing method of a fan according to another embodiment of the present disclosure; and

[0038] FIG. **9** is a flow chart illustrating a manufacturing method of a fan according to further another embodiment of the present disclosure.

DETAILED DESCRIPTION OF EMBODIMENTS

[0039] The present disclosure will now be described more specifically with reference to the following embodiments. It is to be noted that the following descriptions of embodiments of this disclosure are presented herein for purpose of illustration and description only. It is not intended to be exhaustive or to be limited to the precise form disclosed.

[0040] Please refer to FIG. **1** which is a schematic view showing a fan according to a first embodiment of the present disclosure. A fan **10** includes a hub **11**, a plurality of blades **12** and a mute ring **13**. Each blade **12** has a first terminal **121** and a second terminal **122**. The first terminal **121** is annularly disposed at a circumference of the hub **11**, and the mute ring **13** is disposed at respective second terminals of the blades **12**. In this embodiment, the blades **12** and the mute ring **13** both are made of metal material, such as stainless steel, titanium, aluminum alloy etc., but not limited thereto, and other suitable metal materials also can be adopted in accordance with the practical requirements. In other words, in this embodiment, the purpose of improving the performance of the fan **10** is achieved by adopting a mute ring **13** which is made of metal material to reduce the thickness thereof.

[0041] The combination of the blades **12** and the mute ring **13** can be achieved by for example soldering, but not limited thereto. When the soldering process is employed, other than directly soldering the mute ring **13** and the blades **12** which are attached together, a structural cooperation therebetween also can be adopted for further smoothing the soldering process. For example, a portion of the second terminal **122** of each blade **12** can be laterally bent to form a soldering

surface (not shown), namely, each blade **12** can have a portion of the second terminal **122** at a position corresponding to where the mute ring **13** is disposed be bent to form the soldering surface perpendicular to the surface of the blade **12**, and the mute ring **13** can be formed to have openings (not shown) corresponding to each soldering surface of respective blades **12**, so that when the blades **12** and the mute ring **13** are attached together, the combination therebetween can be achieved by filling solders in the openings and then proceeding the soldering process, or by directly performing the soldering process in the openings.
[0042] In other embodiments, each blade **12** and the mute ring **13** are implemented to respectively employ a first engaging structure (not shown) and a second engaging structure (not shown) for further ensuring a stable combination therebetween. For example, the mute ring **13** can be implemented to have notches for inserting the blades **12** therein; or each blade **12** can be implemented to have a recess for receiving the mute ring **13**; or each blade **12** can be implemented to have an engaging element and the mute ring **13** can be implemented to have receiving portions corresponding thereto, so the combination therebetween can be ensured through penetrating the engaging elements into the receiving portions. Therefore, various kinds of matched engaging structures can be adopted without limitation.
[0043] FIG. **2** is a comparison table showing data of the fan adopting the metal mute ring according to the present disclosure and a control fan with non-metal mute ring. As shown, it is known that compared with the fan adopting a non-metal mute ring with thicker thickness (0.5 mm) which provides an air flow of about 3.65 CFM, the fan of the present disclosure adopting the metal mute ring with thinner thickness (0.2 mm) can provide an air flow of about 4.03 CFM, in which the performance is significantly improved at least 10%. This also proves that the expected performance improvement of fan indeed can be achieved through reducing the thickness of the mute ring by changing the material thereof.
[0044] Please refer to FIG. **3**A to FIG. **3**B, FIG. **4**A to FIG. **4**B and FIG. **5**. FIG. **3**A is a schematic view showing a fan according to a second embodiment of the present disclosure, FIG. **3**B is a schematic view showing the fan according to the second embodiment of the present disclosure from another view angle, FIG. **4**A is an enlarged view of square A in FIG. **3**A, FIG. **4**B is an enlarged view of square B in FIG. **3**B, and FIG. **5** is a top view of the fan according to the second embodiment of the present disclosure. A fan **20** includes a hub **21**, a plurality of first blades **22**, a mute ring **23** and a plurality of second blades **24**. Compared with the fan shown in FIG. **1**, the fan **20** further includes the plurality of second blades **24** connected to the mute ring **23**. More specifically, in this embodiment, the first blades **221** are disposed at a circumference of the hub **21** through respective first terminals **221** thereof, and the second blades **24** are disposed at respective second terminals **222** of the first blades **22** through the mute ring **23**, and further, the second blades **24** and the first blades **22** are alternately arranged. That is, one second blade **24** is disposed between every two adjacent first blades **22**, and thus, the numbers of the first blades **22** and the second blades **24** are identical. Under this configuration, compared with a fan with the first blades **22** only, the disposition of the second blades **24** doubles the total number of blades of the fan, which accordingly improves the performance of the fan.
[0045] In particular, the second blades **24** are only disposed at the second terminals **222** of the first blades **22**, and there has no additional blade disposed at the first terminals **221** of the first blades **22**. In other words, the length of each second blade **24** is smaller than the length of each first blade **22**. Namely, as shown in FIG. **5**, a distance D**2** between a third terminal **241** of each second blade **24** toward the hub **21** and a fourth terminal **241** opposite to the third terminal **241** is less than a perpendicular distance DI between the second terminal **222** of each first blade **22** and the circumference of the hub **21**. Through this design, the second blades **24** can be fixed through the mute sing **23**, so that the total number of blades of the fan can be increased and the performance of the fan also can be improved without altering the structures of the hub **21** and the first blades **22**. Here, the length of the second blade **24**, namely, the distance D**2** between the third terminal **241**

and the fourth terminal **242**, can be varied in accordance with the practical situations without limitation.

[0046] More specifically, on the premise that there is limitation on the number of the first blades **22** that can be disposed at the circumference of the hub **21**, by utilizing the larger circumference at the second terminals **222** of the first blades **22** to provide more installation spaces and employing the mute ring **23** at the second terminals **222** for disposing the second blades **24**, the fan of the present disclosure achieves the purpose of increasing the total number of blades without the need of altering the existing architecture thereof, which provides a novel structure capable of improving the performance of the fan.

[0047] In an embodiment, the first blades **22** are made of metal material, such as stainless steel, titanium, aluminum alloy etc., but not limited thereto, and other suitable metal materials also can be adopted in accordance with the practical requirements. On the other hand, the second blades **24** are made of plastic material, such as LCP, PPE, PBT etc., but not limited thereto, and other suitable plastic materials also can be adopted in accordance with the practical requirements. Through the cooperation between the metal first blades **22** and the plastic second blades **24**, the fan of the present disclosure owns advantages of both, so that the performance can be significantly improved, compared to fans with conventional blade structure.

[0048] In another embodiment, the first blades **22** are made of plastic material to cooperate with plastic mute ring **23** and plastic second blades **24**. In this configuration, the total number of blades is doubled, so that the purpose of improving the performance of the fan also can be achieved.

[0049] Please refer to FIG. **6** which is a graph comparing performances of a fan adopting metal first blades and plastic second blades and a fan adapting metal blades only. As shown, through increasing the plastic second blades to alternately arrange among the metal first blades, the performance of fan can be improved at least about 15% to 20%. This proves that increasing the number of blades through the way disclosed in the present disclosure can indeed achieve the expected effect of improving the performance of fan.

[0050] The combination of heterogeneous materials between the metal first blades **22** and the plastic second blades **24** could be achieved by the following methods. Notably, the methods described below are only exemplary, and the scope of the present disclosure is not limited thereby.

[0051] A manufacturing method according to an embodiment of the present disclosure is shown in FIG. **7**. In this embodiment, the mute ring **23** is made of plastic material. In step S**100**, the hub **21** and the first blades **22** are combined to form a fan main body. In step S**102**, the fan main body is positioned in a tool. In step S**104**, an injection molding process is performed to form the mute ring **23** at the second terminals **222** of the first blades **22** and also form the second blades **24** on the mute ring **23**, thereby alternately arranging each second blade **24** between every two adjacent first blades **22**.

[0052] More specifically, in this method, the hub **21** and the metal first blades **22** are combined together to form the fan main body first, and then through performing the secondary processing, not only the formations of the plastic mute ring **23** and the plastic second blades **24** are achieved, but also the combination between the plastic mute ring **23** and the metal first blades **22** is simultaneously completed, thereby forming the mute ring **23** and the second blades **24** which are both made of plastic material at the second terminals **222** of the metal first blades **22**. In an embodiment, the mute ring **23** and the second blades **24** are integrally formed.

[0053] Another manufacturing method according to another embodiment of the present disclosure is shown in FIG. **8**. In this embodiment, the mute ring **23** includes a metal base and a plastic outer portion. In step S**200**, the hub **21** and the first blades **22** are combined to form a fan main body. In step S**202**, the fan main body and the metal base of the mute ring **23** are respectively positioned in a tool and the metal base is located at the second terminals **222** of the metal first blades **22**. In step S**204**, an injection molding process is performed to form the plastic outer portion of the mute ring **23** and also form the plastic second blades **24** among the metal first blades **22**, thereby completing the configuration that the metal first blades **22** and the plastic second blades **24** are alternately arranged.

[0054] More specifically, in this method, the hub **21** and the metal first blades **22** are combined together to form the fan main body first, and then through performing the secondary processing, not only the metal base is covered by the plastic material to form the mute ring **23**, but also the formation of the second blades **24** and the combination and stabilization between the metal base and the metal first blades **24** through the plastic material are simultaneously completed, thereby forming the mute ring **23** and the plastic second blades **24** at the second terminals **222** of the metal first blades **22**. In an embodiment, the plastic outer portion of the mute ring **23** and the second blades **24** are integrally formed.

[0055] Further another manufacturing method according to another embodiment of the present disclosure is shown in FIG. **9**. In this embodiment, the mute ring **23** includes a metal base and a plastic outer portion. In step S**300**, the hub **21** and the first blades **22** are combined to form a fan main body. In step S**302**, the metal base of the mute ring **23** is connected to the second terminals **222** of the metal first blades **22**, for example, through the soldering process described above. In step S**304**, an injection molding process is performed to form the plastic outer portion of the mute ring **23** and also form the plastic second blades **24** among the metal first blades **22**, thereby completing the structure that the metal first blades **22** and the plastic second blades **24** are alternately arranged.

[0056] More specifically, in this method, the hub **21** and the metal first blades **22** are combined together to form the fan main body first. When performing the secondary processing, the metal base is connected to the fan main body first for stabilization, and then, by performing the injection molding process, the plastic outer portion is formed to cover the metal base and the formation of the second blades **24** is also completed, thereby forming the mute ring **23** and the plastic second blades **24** at the second terminals **222** of the metal first blades **22**. In an embodiment, the plastic outer portion of the mute ring **23** and the second blades **24** are integrally formed, and the plastic outer portion is implemented to cover at least a portion of the metal base for ensuring the stabilization of the plastic second blades **24**.

[0057] For the embodiments described above, the plastic material adopted in the injection molding process can be, such as LCP, PPE, PBT etc., but not limited thereto, and other suitable plastic materials also can be adopted in accordance with the practical requirements.

[0058] In addition, the combination between the first blades and the hub also can be varied in accordance with the practical situations. For example, for the hub made of plastic material, the formation of the hub and the combination thereof with the metal first blades can be completed at the same time through the injection molding process; or for the hub made of metal material, the combination between the hub and the metal first blades can be achieved through soldering; or other processes also can be adopted to combine the two, without limitation. Further, the position of the mute ring can be located at the very ends of the second terminals of the first blades (as shown in FIG. **2**A and FIG. **2**B), or at positions near the very ends (as shown in FIG. **1**), which can be varied in accordance with the practical situations.

[0059] In conclusion, the present disclosure provides the mute ring which is made of metal material and has a reduced thickness for cooperating with the metal blades so as to achieve the purpose of improving the performance of fan. Moreover, in the present disclosure, the mute ring is also used as the medium for disposing the additional blades, so that the total number of blades can be doubled for significantly improving the performance of fan without the need of altering the existing structures of the hub and the blades of the fan.

[0060] While the disclosure has been described in terms of what is presently considered to be the most practical embodiments, it is to be understood that the disclosure needs not be limited to the disclosed embodiment. On the contrary, it is intended to cover various modifications and similar arrangements included within the spirit and scope of the appended claims which are to be accorded with the broadest interpretation so as to encompass all such modifications and similar structures.

Claims

1. A fan, comprising: a hub comprising a circumference; a plurality of first blades, each of the plurality of first blades comprising a first terminal disposed at the circumference of the hub and a second terminal opposite to the first terminal; a mute ring disposed at the second terminal of each of the plurality of first blades; and a plurality of second blades disposed at the second terminal of each of the plurality of first blades and connected to the mute ring, wherein the plurality of second blades and the plurality of first blades are alternately arranged.
2. The fan as claimed in claim 1, wherein each of the plurality of second blades comprises a third terminal toward the hub and a fourth terminal opposite to the third terminal, and a distance between the third terminal and the fourth terminal of each of the plurality of second blades is less than a perpendicular distance between the second terminal of each of the plurality of first blades and the circumference.
3. The fan as claimed in claim 1, wherein the plurality of first blades are made of a metal material, and the plurality of second blades are made of a plastic material.
4. The fan as claimed in claim 3, wherein the mute ring is made of the plastic material and is integrally formed with the plurality of second blades.
5. The fan as claimed in claim 3, wherein the mute ring comprises a metal base and a plastic outer portion, and the plastic outer portion is integrally formed with the plurality of second blades.
6. The fan as claimed in claim 5, wherein the mute ring is stabilized at the second terminal of each of the plurality of first blades through the plastic outer portion thereof.
7. The fan as claimed in claim 5, wherein the mute ring is stabilized through connecting the metal base thereof to the second terminal of each of the plurality of first blades.
8. A manufacturing method of a fan, comprising steps of: (a) providing a hub having a circumference; (b) providing a plurality of first blades, each of the plurality of first blades comprising a first terminal and a second terminal opposite to the first terminal, and disposing the first terminal of each of the plurality of the first blades at the circumference of the hub; and (c) performing an injection molding process to form a mute ring and a plurality of second blades at the second terminal of each of the plurality of first blades, wherein the plurality of second blades and the plurality of first blades are alternately arranged.
9. The manufacturing method of the fan as claimed in claim 8, wherein each of the plurality of second blades comprises a third terminal toward the hub and a fourth terminal opposite to the third terminal, and a distance between the third terminal and the fourth terminal of each of the plurality of second blades is less than a perpendicular distance between the second terminal of each of the plurality of first blades and the circumference.
10. The manufacturing method of the fan as claimed in claim 8, wherein the plurality of first blades are made of a metal material, and the plurality of second blades are made of a plastic material.
11. The manufacturing method of the fan as claimed in claim 10, wherein the mute ring is made of the plastic material and is integrally formed with the plurality of second blades.
12. The manufacturing method of the fan as claimed in claim 10, wherein the mute ring comprises a metal base and a plastic outer portion.
13. The manufacturing method of the fan as claimed in claim 12, wherein the step (c) further comprises steps of: (c1) disposing the metal base at the second terminal of each of the plurality of first blades; and (c2) performing the injection molding process to form the plastic outer portion and the plurality of second blades.
14. The manufacturing method of the fan as claimed in claim 13, wherein the step (c1) further comprises: positioning the metal base at the second terminal of each of the plurality of first blades

through a tool.

15. The manufacturing method of the fan as claimed in claim 13, wherein the step (c1) further comprises: connecting the metal base to the second terminal of each of the plurality of first blades.

16. A fan, comprising: a hub comprising a circumference; a plurality of first blades made of a metal material, wherein each of the plurality of first blades comprises a first terminal disposed at the circumference of the hub and a second terminal opposite to the first terminal; and a mute ring made of a plastic material and disposed at the second terminal of each of the plurality of first blades.

17. The fan as claimed in claim 16, wherein the second terminal of each of the plurality of first blades is bent to form a soldering surface, the mute ring comprises a plurality of openings respectively corresponding to the soldering surface of the second terminal of each of the plurality of first blades, and the mute ring is fixed through soldering the soldering surface of the second terminal of each of the plurality of first blades respectively with each of the plurality of openings.

18. The fan as claimed in claim 16, wherein the second terminal of each of the plurality of first blades comprises a first engaging structure, the mute ring comprises a plurality of second engaging structures respectively corresponding to the first engaging structure of the second terminal of each of the plurality of first blades, and the mute ring is fixed through engaging the plurality of second engaging structures respectively with the first engaging structure of the second terminal of each of the plurality of first blades.